

Melt Grafting of Glycidyl Methacrylate Onto Polypropylene and Reactive Compatibilization of Rubber Toughened Polypropylene

LI-FENG CHEN,[†] BETTY WONG, and W. E. BAKER*

Department of Chemistry
Queen's University
Kingston, Ontario, Canada, K7L 3N6

Free radical melt grafting of glycidyl methacrylate (GMA) onto polypropylene (PP) was studied. The extent of GMA grafting and the molecular weight of the functionalized PP copolymers were controlled by carefully manipulating various reaction factors, such as monomer concentration, initiator concentration, reaction temperature, and molecular weight of the starting PP homopolymer. The use of a second monomer, styrene, in the grafting process helped to increase GMA grafting further and reduce chain scission. The GMA modified PP copolymer was found to be able to reactively compatibilize PP/acrylonitrile-co-butadiene-co-acrylic acid rubber (NBR) blends. Up to an eight-fold increase in the impact energy of the PP/NBR blend was obtained. The compatibilizing capacities of the reactive copolymers, in terms of impact energy improvement of the PP/NBR blend, were found not to be exclusively dependent on the total concentration of reactive functionalities in the matrix of the blend. The characteristics of the reactive copolymers, i.e., the extent of functionalization and the molecular weight, were found to have significant influences on the compatibilizing capacity. A large amount of moderately functionalized copolymer offers better compatibilization performance than a small amount of highly functionalized copolymer. A significant drop in impact energy was observed with declining molecular weight of the copolymer.

INTRODUCTION

Polymer blends have become a versatile and economically viable solution to the needs of industry for improving the performance of commodity and engineering plastics (1). However, many polymers are immiscible with each other and are phase separated when blended, which is due to the low entropy of mixing of long polymer chains. This phenomenon provides the advantage of maintaining some of the individual properties of the blend's components. However, coarse and unstable morphology, and lack of interfacial adhesion in the solid state lead to poor mechanical performance, particularly for those related to ductility. Ide and Hasegawa (2), and Heikens *et al.* (3, 4) were among the early researchers who observed that addition of small amount of a third component, to a binary blend, stabilizes the phase morphology. Reductions in the dispersed phase particle size went hand-

in-hand with enhanced mechanical performance of these immiscible blends.

A common compatibilization strategy was to introduce a block copolymer, with the blocks chemically identical to, or miscible with, the respective blend components. Compatibilization can also be achieved by the so-called reactive compatibilization. In many situations, one or both of the polymer phases do not possess any functional groups; however, reactive groups can be incorporated into these phases by introducing a functionalized polymer with which they are miscible. Graft or block copolymers acting as compatibilizers in polymer blends can then be formed *in situ* through covalent or ionic bonding at the interface during melt blending of suitably functionalized polymers (5-9).

Many of the functionalized reactive polymers are prepared by a melt grafting process because of the relative simplicity and cost efficiency of the process. Polypropylene (PP) is a widely used commodity polymer and its blends with various other polymers have been the target of extensive research. Studies on PP grafted with maleic anhydride (MA) (2, 10-12) and

* To whom correspondence should be addressed.

† Horizon Plastics Co. Ltd., Cobourg, Ontario.

carboxylic acid (13, 14), and the subsequent compatibilization of PP blends are frequently found in the literature. However, relatively fewer papers are available on grafting of PP with other reactive functionalities. Liu *et al.* have successfully grafted a vinyl oxazoline, and amino-, hydroxy-, and glycidyl methacrylate functionalities onto PP (15). Glycidyl methacrylate (GMA) contains an epoxide, which is highly electrophilic and capable of reacting with a variety of functional groups including weakly nucleophilic species, such as carboxylic acids. Sun *et al.* have also studied melt grafting of GMA onto PP with styrene added as a comonomer to reduce β -scission of the PP backbone during the peroxide initiated grafting reaction and to enhance the grafting efficiency of GMA (16).

Various polymer blends have been compatibilized with GMA containing copolymers. Liu *et al.* found that PP-g-GMA is effective in compatibilizing PP/acrylonitrile-butadiene-acrylic acid rubber (NBR) blends (15, 17), in terms of improved impact strength and finer morphology. Chang and coworkers (18–20) have used styrene-glycidyl methacrylate copolymers (SG) prepared through copolymerization, as reactive compatibilizers in polystyrene (PS) blends with poly(ethylene terephthalate) (PET), poly(butylene terephthalate) (PBT), and polyamide-6,6. Tsai *et al.* prepared PS-g-GMA copolymers, which were used to compatibilize PS/PET blends (21). Holsti-Miettinen *et al.* compatibilized polyamide-6/PP blends with an ethylene-ethyl acrylate-GMA terpolymer (22). GMA functionalized ethylene-propylene rubber was used by Vongpanish *et al.* to toughen PET (23). The reaction involved in these blend compatibilization studies is between the epoxy and carboxylic acid groups, which results in the formation of ester linkages between the two polymer phases.

The structural characteristics of compatibilizers are key factors in determining the blend performance. Most of these structure/performance studies have been carried out using block copolymers. Fayt and Teyssié (24) added poly(hydrogenated butadiene-styrene) block copolymers to polyethylene/polystyrene blends and found that the mechanical performance is strikingly dependent on the molecular characteristics of the copolymers. In a series of studies carried out by Kramer *et al.* (25–27) and Brown *et al.* (25, 28, 29), it was found that the interfacial adhesion and in turn the fracture mechanism is dependent on the degree of polymerization of each block, the areal density of block copolymer chains at the interface, and the average degree of polymerization between entanglements of the two homopolymers. Char *et al.* performed a similar study on the interfacial adhesion between amine-terminated poly(methyl methacrylate) and poly(phenylene ether) that were compatibilized with a reactive polystyrene copolymer containing anhydride (30). They found that the total amount of functional groups seemed to be a dominant factor determining the amount of copolymer formed at the interface. Duvall *et al.* compared two reactive PP's, one of low MA content (LAC) and one of high MA content (HAC), in

compatibilizing PP/polyamide-6,6 blends (31–33). A much higher maximum fracture strain was achieved with LAC than with HAC in PP rich blends. But in polyamide rich blends, similar effects on particle size and tensile properties were observed for both copolymers.

In this work, peroxide initiated, free radical melt grafting of GMA onto PP was studied. The emphasis was on preparing a family of PP-g-GMA compatibilizers with a wide range of tailored, reactive group contents and molecular weights. NBR rubber containing acrylic acid was blended with PP to enhance the toughness of the relatively brittle PP. The GMA modified copolymers, with tailored features, were added to the PP phase in order to examine the effects of the degree of grafting and copolymer molecular weight on the impact properties of PP/NBR blends.

EXPERIMENTAL

Free Radical Melt Grafting of GMA onto PP

The PP resins used were nonreactive homopolymers supplied by Shell. Selected grades of PP ranging from GE6100 with low melt flow index (MFI) of 0.8 g/10 min (ASTM D1238, 230°C and 2.16 kg) to WM6100 with a high MFI of 30 g/10 min were used in the grafting and blending studies. GMA monomer (97%, MW 142, Aldrich) inhibited with 50 ppm *p*-methoxy phenol (MEHQ), styrene monomer (99%, MW 104, Aldrich) inhibited with 15 ppm 4-*tert*-butylcatechol, and 2,5-dimethyl-2,5-di-(*t*-butylperoxy) hexyne-3 (Lupersol 130, MW 286, Elf Atochem) peroxide initiator were used as received. The Lupersol 130 initiator has a half-life of 8 min at 170°C, 3.3 min at 180°C, and 1.5 min at 190°C, respectively, in dodecane (34).

The grafting reactions were carried out in a Banbury-like batch mixer (Haake Rheomix 600, 50 mL³) attached to a Haake Rheocord System 40. The mixing torque and melt temperature were monitored on-line and recorded. PP pellets, monomer(s) (2–16 wt%), and initiator (0.2–1.0 wt%) were dry-blended before being charged into the mixer. The rotor speed and mixing time were fixed at 100 RPM and 7 min. The melt temperature was selected between 170°C and 190°C, usually 180°C. In some of the grafting experiments styrene was added as a comonomer to reduce β -scission and enhance the grafting of GMA (16). Different grades of PP homopolymer were used in the grafting process, thus providing one of the factors to control the molecular weight (or MFI) of the final graft copolymer.

Purification and Characterization of PP-g-GMA

The crude graft copolymer was dissolved in refluxing toluene (~4 g per 100 mL solvent) followed by precipitation with ten volumes of acetone. The unreacted monomer(s), GMA homopolymer, and GMA-styrene copolymer remained soluble in acetone and were separated out from the grafted PP. The precipitated graft modified PP was filtered, washed, and then vac-

uum dried at 80°C for at least 12 hours. FT-IR measurements showed that DG level remain unaltered upon further rounds of purification; therefore, one purification step was sufficient for removal of all the residual impurities.

The degree of grafting (DG) was defined as the weight percentage of the reactive component bound onto the graft copolymer. The purified copolymers were compressed into thin films, 0.1–0.2 mm in thickness, at 180°C. Absorbance spectra of the copolymer films were acquired using a Bomem MB-120 FT-IR spectrometer (Fig. 1). The carbonyl peak at 1725 cm^{-1} is characteristic of GMA and the peak at 701 cm^{-1} for styrene was selected. A calibration curve was established between the DG and the peak intensities, which was corrected for the actual film thickness. Another calibration method based on absorbance peak ratios with the PP peak at 2723 cm^{-1} was developed by Sun *et al.* (16). Good correlation exists between these two calibration methods.

Because of the degradation of PP caused by β -scission during the melt grafting process, the MFIs of the purified copolymers were too high to be measured accurately at 230°C. Therefore, MFI measurements were carried out at 190°C with a 2.16 kg weight, rather than at the standard ASTM condition for PP, for comparison purposes. Unless otherwise stated, the MFI values presented in this paper, including those for the PP homopolymers, were measured under this condition.

The viscosity average molecular weights, $\bar{M}_v$, of some of the copolymers were measured using an Ubbelohde viscometer with a built-in filter. Decahydronaphthalene was used as the solvent. Viscometry measurements were carried out at 135°C in a silicon oil bath. The Mark-Houwink constants for this particular solvent-temperature system are, $K = 0.0158 \text{ mL/g}$ and $\alpha = 0.77$ (35). The MFI values were found to

correlate with the actual molecular weight (Table 1); therefore, molecular weight measurements were carried out only on selected PP-g-GMA samples. Free radical induced β -scission of PP has been found to reduce first the polydispersity ($\bar{M}_w/\bar{M}_n$), which eventually stabilizes at a constant value (36). PP-g-GMA copolymers made by a similar melt grafting process were found to have polydispersity values around 3.1 (16). Since $\bar{M}_v$ should be close to $\bar{M}_w$ for polymers with narrow molecular weight distributions, $\bar{M}_n$ was estimated to be 0.32 $\bar{M}_v$ for comparison purposes in explaining the results from the reactive compatibilization studies.

Reactive Compatibilization of PP/NBR Blends

The NBR rubber used was Krynac X7.40, obtained from Bayer Rubber Inc., and has a $\bar{M}_w$ of 310,000 g/mol and Mooney viscosity of 50. It is a terpolymer of acrylonitrile-butadiene-acrylic acid, with 7 wt% acrylic acid content. Acrylic acid is the reactive functionality in the rubber and is in large excess compared to the epoxy functionalities added to the blend matrix. The NBR rubber concentration in the blend was kept constant at 20 wt%. The other 80% of the blend consisted primarily of the PP homopolymer and varying small amounts of PP-g-GMA copolymer.

In order to separate out the effects of the reactive content from those of the compatibilizer molecular weight, only PP-g-GMA copolymers with similar MFIs to those of the PP homopolymers were used in the reactive blending studies. This rule was bypassed only when the molecular weight of the graft copolymer was being studied, for which a family of PP-g-GMA's with different MWs, but similar DGs of GMA, were selected.

The blending was also carried out in the Haake batch mixer. The NBR rubber was preblended with 0.5 wt% antioxidant (Wingstay LHLS, hindered phenol

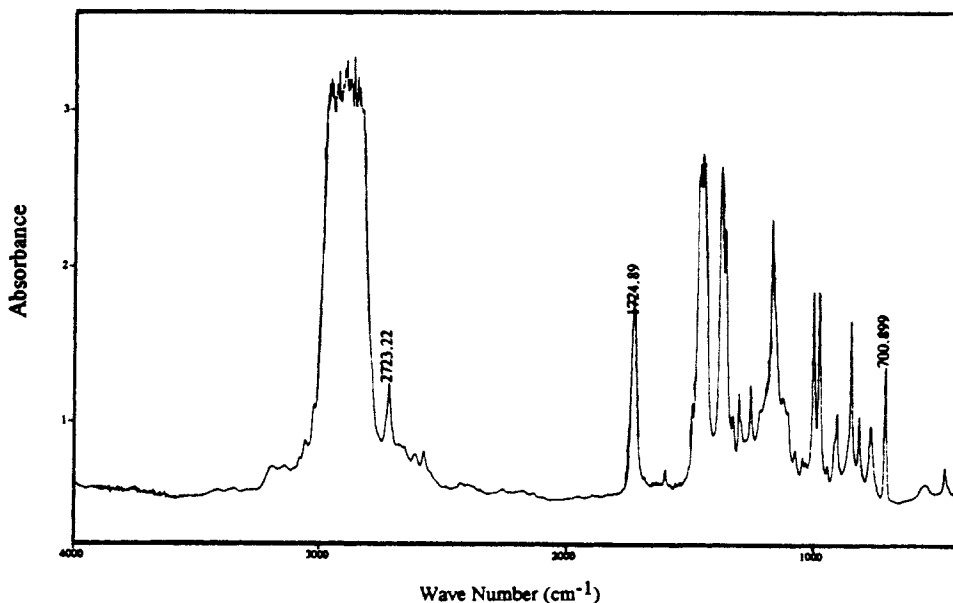


Fig. 1. FT-IR absorbance spectrum of a PP-g-GMA-g-styrene copolymer.

Table 1. Correlation Between Molecular Weight (MW) and Melt Flow Index (MFI) for Some PP Homopolymers and Reactive Copolymers. $\bar{m}_w$ Data Provided by Shell. The MFI Was Measured at 190°C, 2.16 kg.

Polymer	Molecular Weight (kg/mol)	Type of MW	Melt Flow Index (g/10 min, at 190°C)
PP (GE 6100)	650	$\bar{M}_w$	0.3
PP (KY 6100)	410	$\bar{M}_w$	1.2
PP (SY 6100)	300	$\bar{M}_w$	5.3
PP (WM 6100)	145	$\bar{M}_v$	13
PP-g-GMA (sample #1)	117	$\bar{M}_v$	17
PP-g-GMA (sample #2)	85	$\bar{M}_v$	25

stabilizer from Goodyear) at 80°C, 100 RPM for 4 min in a separate run in order to minimize oxidation of the rubber at high processing temperatures and long mixing times.

The mixing temperature was kept at 190°C and the mixing speed at 100 RPM for all the blends. The PP homopolymer was melt blended with the PP-g-GMA copolymer for 3 min to ensure good distributive mixing of the compatibilizer in the PP matrix, before the rubber was added. The blend was then processed for an additional 7 min.

PP homopolymer and PP-g-GMA blends (with identical compositions to the matrices of the corresponding compatibilized blends) and nonreactive PP/NBR blends, were made as control experiments. Also, PP homopolymers were processed alone under the same mixing conditions to ensure that all the materials had the same thermal history.

Impact Energy Measurements

Unnotched disk shaped specimens with a diameter of 37 mm and a thickness of 3.2 mm were injection molded at 190°C. The test specimens were left at room temperature for at least two days before testing to allow some extent of self-annealing.

The impact energy data was obtained using a Rheometrics RDT-5000 instrumented drop weight impact tester. The diameters of the probe and supporting ring were 12.7 mm and 25.4 mm, respectively. The weight of the probe was 3.75 kg. All testing was carried out at room temperature. The probe was dropped from a height of 0.97 m at an impact speed of 7.6 m/s. Ten specimens were tested for each blend sample. The impact energy was calculated as the area below the force vs. displacement curve.

The blend morphologies were examined using a JEOL JSM 840 scanning electron microscope. The samples were prepared by cryogenically fracturing injection molded tensile bars. The test specimens were immersed in 1M chromic acid at 70°C for 10 min, to remove the dispersed phase (NBR) from the surface, and vacuum coated with gold prior to examining.

RESULTS AND DISCUSSION

Free Radical Melt Grafting Reactions on PP

The free radical grafting reaction starts with thermal decomposition of peroxide initiator into free radicals, which are capable of abstracting hydrogen atoms from the PP molecules. PP macroradicals are highly reactive species that will react with monomer(s) present in the melt system. However, the macroradicals also undergo chain-scission (β -scission) because of the presence of methyl side groups. Hence, there exists a competition between the grafting of monomer molecules onto PP and β -scission of PP molecules themselves, which can lead to the formation of block copolymers. Degradation of PP during the grafting process was observed and is reflected by the significant increase in the MFI values of the resulting copolymers. The degradation can be controlled, however, by varying the monomer concentration, initiator concentration, processing temperature, etc., as will be discussed later.

The FT-IR absorbance bands at 944 cm^{-1} and 850 cm^{-1} (39), characteristic of the epoxy group in the grafted copolymer, are relatively weak compared to the carbonyl peak and are overlapped by PP bands; hence, they can only be seen after the spectrum of the pure PP is subtracted. The FT-IR subtraction spectrum of the grafted copolymer showed that the epoxy group peaks from GMA were present and no -OH broad band at around 3500 cm^{-1} was observed. This indicates that the epoxy rings remained intact after the grafting process and are available for reaction with acid groups in the PP/NBR blends later. The disappearance of epoxy bands has actually been utilized to study the reaction between epoxy and carboxylic acid groups (21, 40, 41). It was found that such a reaction is rapid, and the first order rate constant for this reaction is 0.91 min^{-1} (41).

The grafting of GMA appeared not to be significantly affected by reaction temperature in the range of 170–200°C. A less than thorough explanation would be that most of the initiator is able to decompose and react with the PP effectively in this temperature range and mixing time. However, PP degradation increased significantly with increasing temperature, therefore, 180°C seemed to be the optimal grafting temperature for this system and became the chosen temperature, unless very low MW copolymers were needed.

To better understand the effects of GMA grafted PP copolymers as blend compatibilizers, it is desirable to know the average length of the GMA grafts on the PP molecules. However, measuring the graft chain length directly is a rather difficult task, if possible at all, so rough estimates were made for typical grafting trials. For example, 45 g PP, 5 g GMA, and 0.3 g Lupersol 130 yielded about 40 g of PP-g-GMA with 2.1 wt% GMA. The reaction was carried out at 180°C, and a mixing time of 7 min ensured almost total decomposition of all the initiator. Each initiator molecule decomposes into four free radicals. Based on the assumption that

all the radicals reacted with the PP and all the resulting macroradicals reacted with GMA monomer, the average graft chain consists of one and one half GMA units. However, some of the free radicals are consumed by GMA homopolymerization, which lowers the number of free radicals effective in grafting GMA onto PP. The grafts are not expected to be long, since the grafting reaction is also limited by the slow diffusion of GMA monomer in the viscous melt and because of the evidence for a ceiling temperature of about 165°C in the DMAEMA system (37). Therefore, a conservative estimate of the GMA graft chain length is 3–4 GMA units on average. This is not inconsistent with the findings of Wong Shing *et al.* (37, 38) who, working with 2-(dimethylamino) ethyl methacrylate (DMAEMA) grafted onto squalane, found that the grafts consisted of 1–2 DMAEMA units on average.

A second method that can be used to obtain information on the graft chain length is to analyze the homopolymer that is formed during the grafting process. The GMA homopolymer and GMA-styrene copolymer that were isolated from the purification process consisted mainly of low molecular weight oligomers (MW below 0.5 kg/mol) with some low molecular weight polymer (4–6 kg/mol). Inferences regarding the graft chains, using information obtained from the analysis of the homopolymer, should be made with care. Wong Shing has suggested that the grafting of DMAEMA onto low MW hydrocarbons and homopolymerization occurred through separate mechanisms. The temperature dependence of the two processes were different; as the temperature was increased, the rate of homopolymerization decreased due the ceiling temperature effect, whereas the rate of grafting was relatively unaffected. However, since it is difficult to produce long chain homopolymers under the reaction conditions, it is likely that the grafted side chains are short.

Single Monomer (GMA) Grafting System

For the GMA grafted PP copolymers, the two parameters believed pertinent to reactive compatibilization of PP/NBR blends have been considered here, namely the GMA content and the molecular weight (represented by its MFI value). The GMA content is represented by the degree of grafting of GMA, DG_g , defined as the weight percentage of GMA grafts in the copolymer. In order to study separately the effects of DG_g and molecular weight, a large family of PP-g-GMA copolymers, or PP-g-GMA-g-styrene copolymers (in the case of dual-monomer grafting system), with different combinations of the two parameters were synthesized by carefully controlling the grafting reaction conditions and the reagent compositions.

The reproducibility of the grafting reaction based on several separate experimental runs at identical reaction conditions was investigated (Table 2). The DG_g of the copolymer was fairly consistent; however, the melt flow index of the copolymer was less reproducible because the molecular weight of the copolymer is quite

Table 2. Reproducibility of the Melt Grafting Reaction. MFI of the Starting PP Homopolymer was 0.3. Starting Initiator Concentration Was 0.67 (g/100 g PP). Initial GMA Monomer Concentration was 11.1 (g/100 g PP).

Experiment #	DG of GMA (wt%)	Melt Flow Index (g/10 min)
1	2.1	12
2	1.9	17
3	2.1	19
average	2.0 ± 0.1	16 ± 3.6

sensitive to the β -scission reaction and melt temperature fluctuations.

The DG_g of PP-g-GMA was found to increase with an increasing initial GMA monomer concentration (Fig. 2). A similar curve was observed for different peroxide initiator concentrations. As mentioned previously, there is a competition between the grafting of GMA monomer and β -scission at the macroradical sites generated by the primary free radicals. Because the total number of available radical sites remains about the same for a particular initiator concentration, it is obvious that increasing GMA monomer concentration would shift the balance towards more extensive grafting. Although the higher GMA monomer concentration does not eliminate the β -scission reaction, it does suppress the degradation of the PP molecules (as reflected by reduced MFI, see Fig. 3). In this Figure, it is evident that substantial β -scission occurs since the starting MFI was 0.3. Hence, control of this reaction was a major issue in preparing these reactive compatibilizers with appropriate molecular weights.

Raising the GMA monomer concentration too high would result in a more rapid increase in the amount of GMA homopolymer formed and unreacted GMA

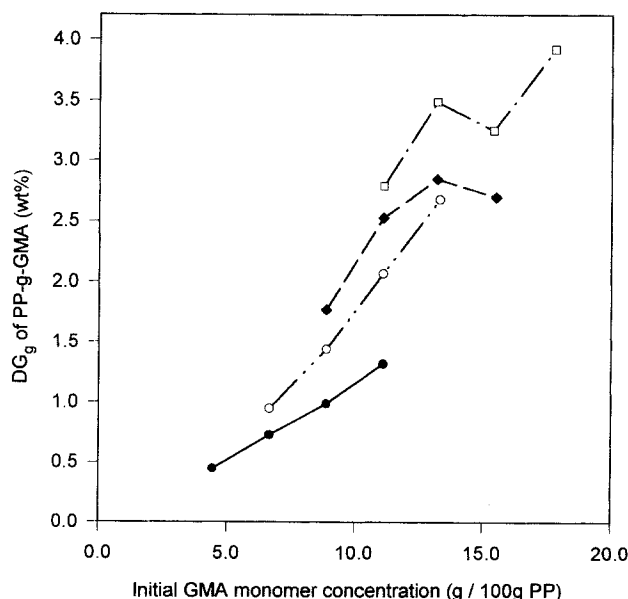


Fig. 2. Effect of initial GMA monomer concentration on DG_g of the PP-g-GMA copolymer. Starting initiator concentration (g / 100 g PP) was: ● 0.44; ○ 0.67; ◆ 0.89; □ 1.11.

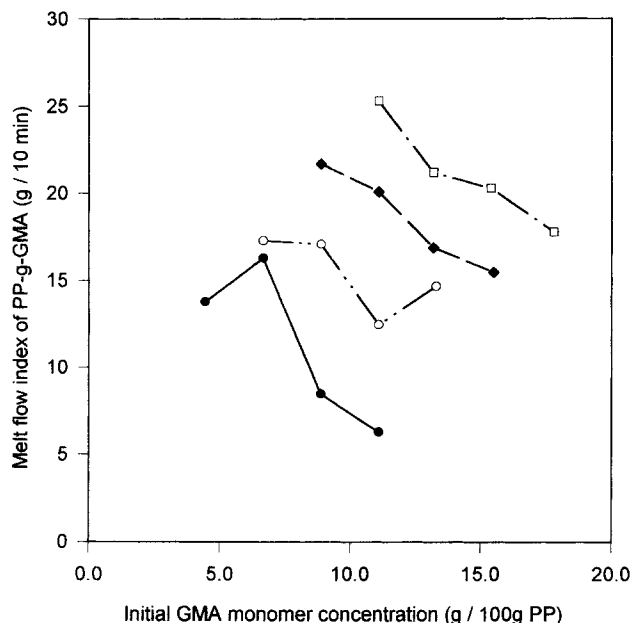


Fig. 3. Effect of initial GMA monomer concentration on melt flow index of the PP-g-GMA copolymer. MFI of starting PP homopolymer was 0.3. Starting initiator concentration (g / 100 g PP) was: ● 0.44; ○ 0.67; ◆ 0.89; □ 1.11.

monomer remaining, reducing the efficiency of the grafting process. Another alternative route to obtain high DG_g PP-g-GMA copolymers is to add more initiator. It is obvious that increasing the initiator concentration generates more radical sites for the grafting reaction, but also for the β -scission reaction. As shown in Fig. 4, at several different GMA monomer concentrations, the DG_g increased with increasing initiator concentration; however, the extent of PP degra-

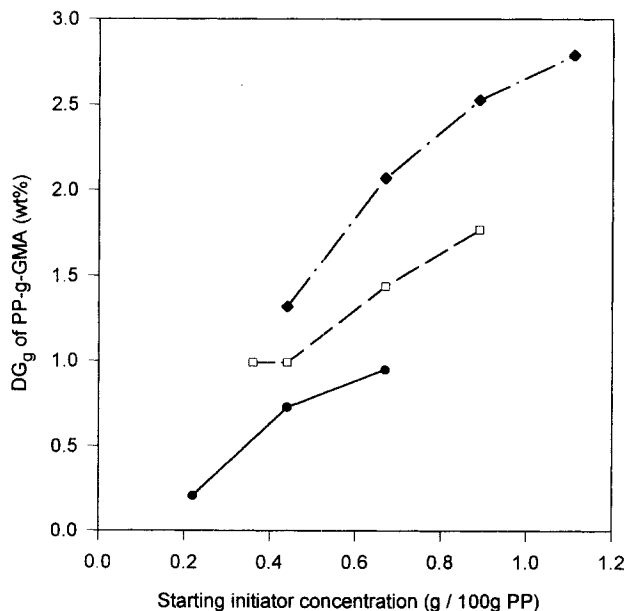


Fig. 4. Effect of starting initiator concentration on DG_g of the PP-g-GMA copolymer. Initial GMA monomer concentration (g / 100 g PP) was: ● 6.7; □ 8.9; ◆ 11.1.

dation was more severe, as noted in Fig. 5. From an optimization standpoint, a balance exists among degree of grafting, chain scission, and homopolymer formation.

Dual-Monomer (Styrene-GMA) Grafting System

In some of the grafting experiments, styrene monomer was also added along with GMA monomer. Addition of styrene monomer reduced the PP degradation and also enhanced the grafting of GMA. The styrene monomer concentrations were set at 0.5, 1 or 1.5 molar ratio to the GMA monomer concentrations added, in order to systematically study the synergistic effects of the dual-monomer feed on the grafting reaction.

The DG_g of GMA of the resulting PP-g-(GMA-styrene) was still dependent on the initial GMA monomer concentration, at various styrene-to-GMA monomer molar ratio (Fig. 6). However, higher levels of GMA were grafted as the addition of styrene monomer enhanced the grafting of GMA (Fig. 7). The synergistic effects are most apparent at 1:1 styrene-to-GMA molar ratio for different GMA monomer concentrations. The DG_g drops slightly as the styrene-to-GMA molar ratio goes up to 1.5, probably because the competition from styrene monomer exceeds its synergistic effect at a high styrene-to-GMA molar ratio. The disadvantage of adding styrene monomer is that styrene is unavoidably grafted onto PP. The content of styrene grafted on the PP copolymers increased with increasing styrene monomer concentration (Fig. 8), but is not strongly affected by the GMA monomer concentration.

The free radical reactivity of styrene towards the PP macroradical is higher than that of GMA (16) when the

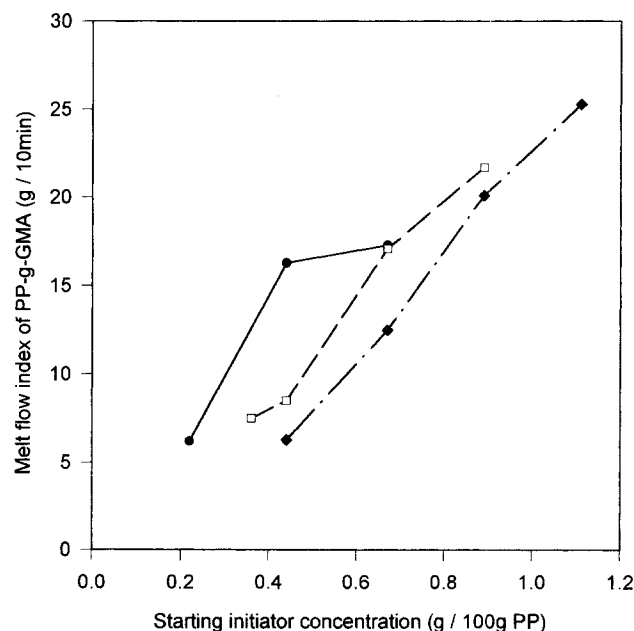


Fig. 5. Effect of starting initiator concentration on melt flow index of the PP-g-GMA copolymer. MFI of the starting PP was 0.3. Initial GMA monomer concentration (g / 100 g PP) was: ● 6.7; □ 8.9; ◆ 11.1.

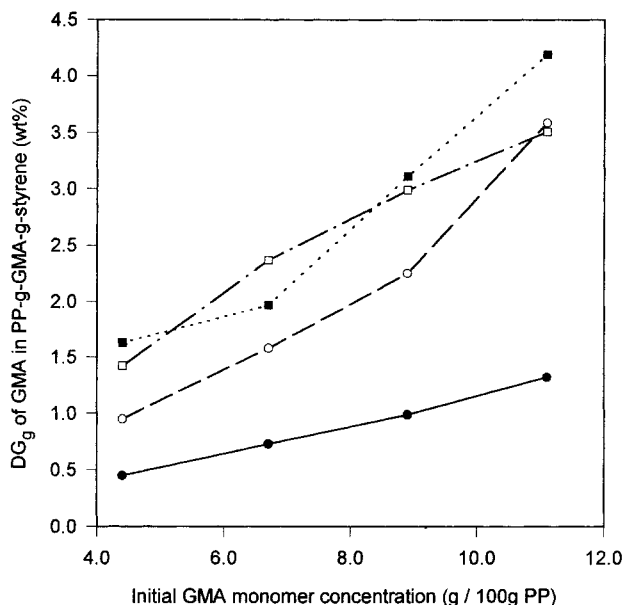


Fig. 6. Effect of initial GMA monomer concentration on DG_g of GMA in the PP-g-GMA-g-styrene copolymer. Starting initiator concentration was 0.44 g/100 g PP. Styrene-to-GMA molar ratio was: ● 0.0 (no styrene); ○ 0.5; ■ 1.0; □ 1.5.

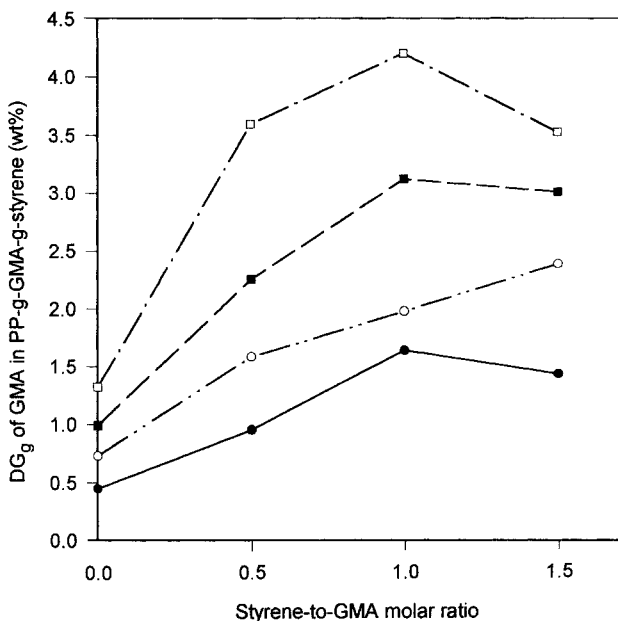


Fig. 7. Effect of styrene-to-GMA molar ratio on the DG_g of GMA in the PP-g-GMA-g-styrene copolymer. Starting initiator concentration was 0.44 g/100 g PP. Initial GMA monomer concentration (g/100 g PP) was: ● 4.4; ○ 6.7; ■ 8.9; □ 11.1.

two monomers are grafted alone. Sun *et al.* proposed that styrene monomer grafts preferentially onto PP first in the dual monomer system, forming more stable styryl macroradicals, which are capable of reacting with GMA. It was also suggested that the graft chain should consist of randomly distributed GMA and styrene units. The rate of reaction of GMA with these styryl radicals is much faster than the rate at which GMA reacts with the PP macroradicals; hence, there is

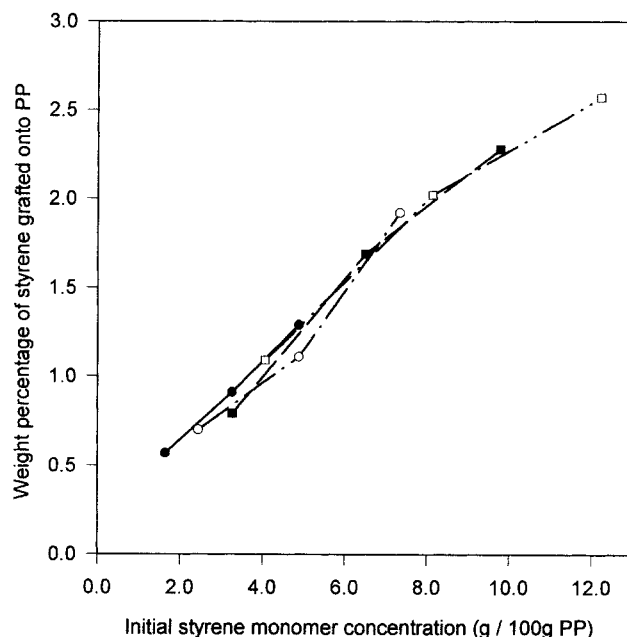


Fig. 8. Effect of initial styrene monomer concentration on weight percentage of grafted styrene in the PP-g-GMA-g-styrene copolymer. Starting initiator concentration was 0.44 g/100 g PP. Initial GMA monomer concentration (g/100 g PP) was: ● 4.4; ○ 6.7; ■ 8.9; □ 11.1.

an increase in the DG of GMA. One of the factors which limits the grafting reaction is the inability of the monomer to reach the macroradical due to the low solubility of the GMA monomer in polypropylene. The addition of styrene as a comonomer may enhance the grafting of GMA in yet another way. As styrene is a good solvent for PP, its presence may create an environment that is more suitable for grafting. The extent of β -scission would also decrease, as more PP macroradicals would be consumed by graft initiation.

Both GMA and styrene monomer can suppress β -scission. The MFI of the graft copolymer decreased with increasing GMA monomer concentration and decreased even more rapidly when progressively more styrene monomer was added (Fig. 9). From the figure, it is apparent that the contribution of styrene in suppressing β -scission was much greater than GMA. This is consistent with the explanation in the previous paragraph. At most GMA monomer concentration levels, only slight changes in MFI were obtained for 0.5 styrene-to-GMA molar ratio (Fig. 10). A sharp drop in MFI was observed when the molar ratio went from 0.5 to 1.0, before leveling off as the molar ratio went from 1.0 to 1.5. Not only does this excess styrene compete more with GMA to inhibit further GMA grafting, but also at a ratio of 1.5, the MFI of the grafted material is at or below that of the starting PP. Possible crosslinking was indicated by an increase in torque after a certain period of melt mixing, and by observing some insoluble gel material during purification. This crosslinking phenomenon will be discussed more extensively in another paper (42). Based on the effects of styrene concentration discussed above, the optimal

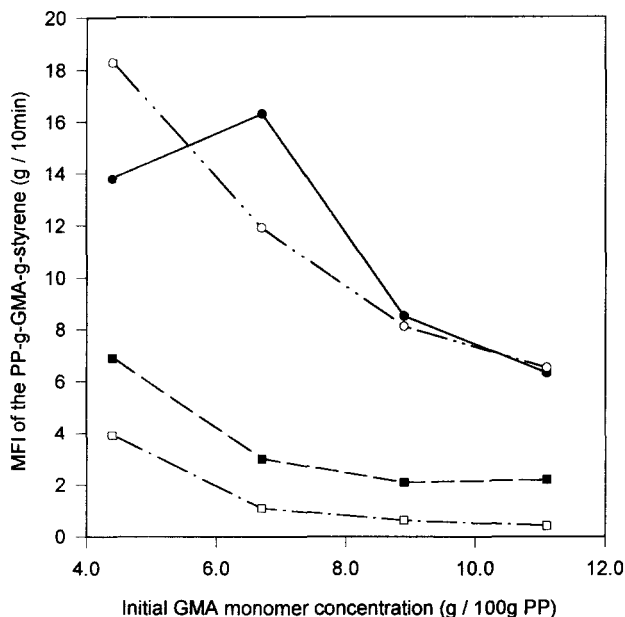


Fig. 9. Effect of initial GMA monomer concentration on melt flow index of the PP-g-GMA-g-styrene copolymer. MFI of starting PP homopolymer was 0.3. Starting Initiator concentration was 0.44 g / 100 g PP. Styrene-to-GMA molar ratio was: ● 0.0 (no styrene); ○ 0.5; ■ 1.0; □ 1.5.

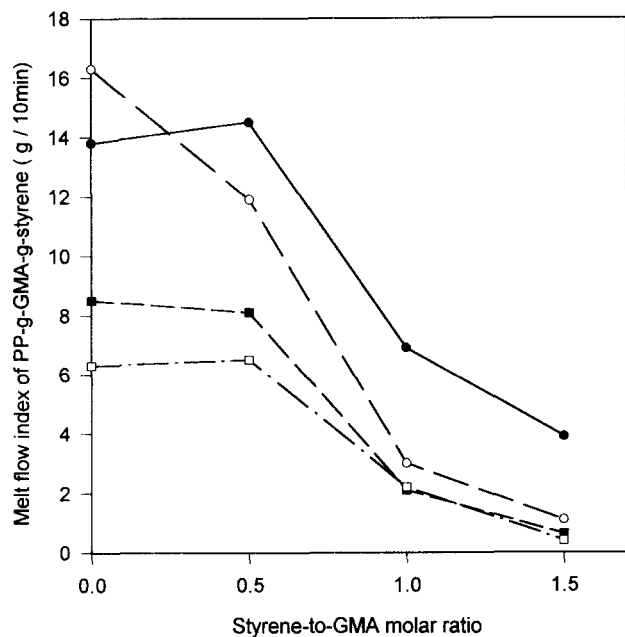


Fig. 10. Effect of styrene-to-GMA molar ratio on the melt flow index of the PP-g-GMA-g-styrene copolymer. MFI of starting PP homopolymer was 0.3. Starting initiator concentration was 0.44 g / 100 g PP. Initial GMA monomer concentration (g / 100 g PP) was: ● 4.4; ○ 6.7; ■ 8.9; □ 11.1.

styrene-to-GMA molar ratio for this grafting system should be 1:1.

Effects of GMA Grafted PP Copolymers on Reactive Compatibilization

The effects of compatibilization can be easily seen by comparing the morphologies of the non reactive

blend to that of a reactive blend (Fig. 11). The blends consist of 80 wt% PP and 20 wt% NBR. Evidence of particle pullout was found for the nonreactive blend (Fig. 11a), indicating that there was little adhesion between the two phases. By comparison, there is little sign of matrix-particle decohesion for the reactive blend (Fig. 11b), indicating that there was increased adhesion between the phases upon the addition of PP-g-GMA. Figure 12 illustrates how the dispersed phase particle size changes with increasing levels of reactivity added to the system. The NBR has been removed from the surface with chromic acid to increase the resolution of the blend components. It can be seen that for the non reactive blend (Fig. 12b), the particle size is large and the dispersion of the NBR in the PP matrix is poor. As the level of PP-g-GMA added to the blend was increased, there was a corresponding decrease in the particle size and the dispersion of the NBR became more homogeneous.

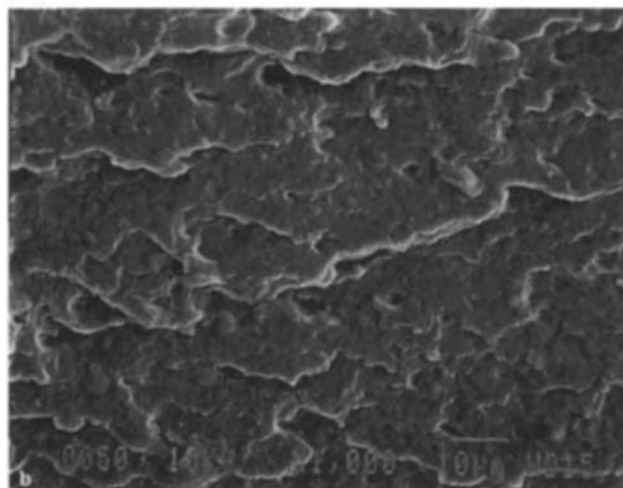
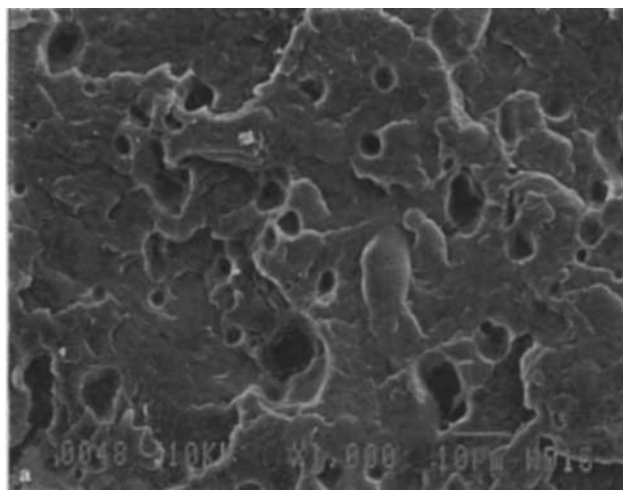


Fig. 11. SEM micrographs of PP/NBR (80:20) blends: a) non-reactive and b) reactive (5 g of PP-g-GMA, DG 1.2 wt%).

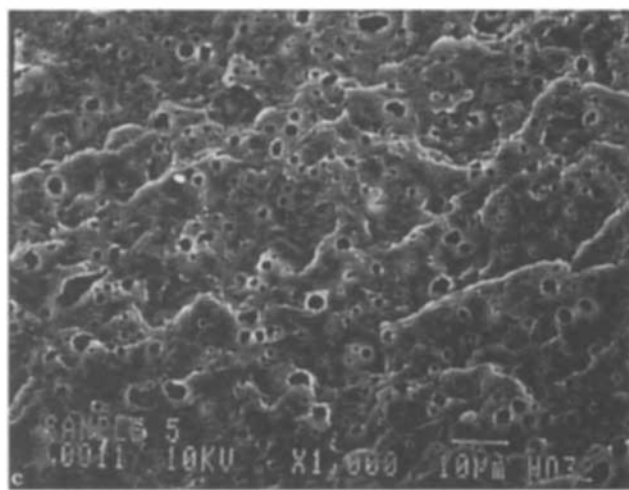
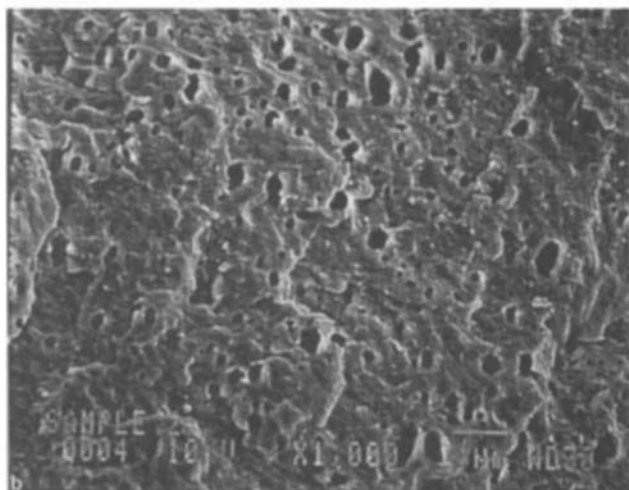
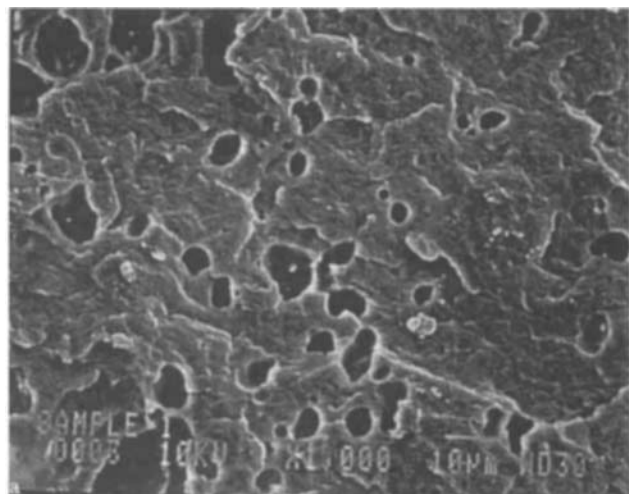


Fig. 12. SEM micrographs of PP/NBR (80:20) blends. The dispersed phase particles were removed from the surface using chromic acid. The DG of the reactive copolymer was 1.2 wt%: a) nonreactive, b) 5 g PP-g-GMA, and c) 7 g of PP-g-GMA.

In the PP/NBR (80/20) blends, NBR rubber was added to enhance the impact energy of the otherwise low toughness PP. However, when only NBR rubber was added without the reactive copolymer, there was essentially little change in the impact energy compared to virgin or processed PP homopolymer (see Table 3). The same trend was observed in earlier work (17). In the PP/NBR blends without the reactive compatibilizer, particle/matrix decohesion prohibits energy from dissipating effectively into the rubber phase. When PP-g-GMA reactive copolymer was progressively added, the impact energy of the PP/NBR blends increased steadily (Fig. 13). Up to an eight fold increase in the impact energy of the PP/NBR blend was achieved compared to that without reactive copolymer.

The PP-g-GMA molecules at the interface react with the carboxylic acid groups on the NBR dispersed phase by ring opening to form ester linkages (21). Meanwhile, the PP-g-GMA molecules remain entangled with other PP molecules in the matrix phase, due to the structural similarity. The reacted PP-g-GMA molecules act as bridges between the matrix phase and the dispersed phase. Therefore, the interfacial adhesion not only depends on the number of interfacial bonds formed with the rubber phases, but also on the effectiveness of the entanglements of the bridging copolymer molecules with other PP molecules in the matrix. The failure mechanism was transformed from brittle fracture to semi-ductile failure, with significant stress whitening observed, once sufficient quantities of reactive copolymers were added. With adequate interfacial adhesion, the rubber particles can act as stress concentrators and are able to absorb more energy, possibly as a result of cavitation of the rubber particles (43).

As the weight percentage of PP-g-GMA copolymer added to the PP matrix progressively increased, the total concentration of reactive groups in the matrix, determined from the DG_g of the copolymer multiplied by its weight percentage in the PP matrix, correspondingly increased. Assuming that a similar proportion of reactive PP molecules manage to get to the interface during the course of blending, as are in the matrix initially, an increasing number of interfacial bonds would be formed. This results in higher impact energies, which is due to enhanced interfacial adhesion (Fig. 13). It was also found that the impact energy of the PP homopolymer remains almost unchanged when blended with varying amounts of PP-g-GMA co-

Table 3. Impact Energies of PP Homopolymer and PP/NBR Blend Without Compatibilizer. Virgin PP Refers to PP (WM 6100) Injection Molded Into Disk Specimen as Received. Processed PP Refers to PP Blended Alone at Exactly the Same Conditions as the PP/NBR Blends.

Polymer	Impact Energy (J)
Virgin PP	0.95
Processed PP	1.16
PP/NBR blend	1.05

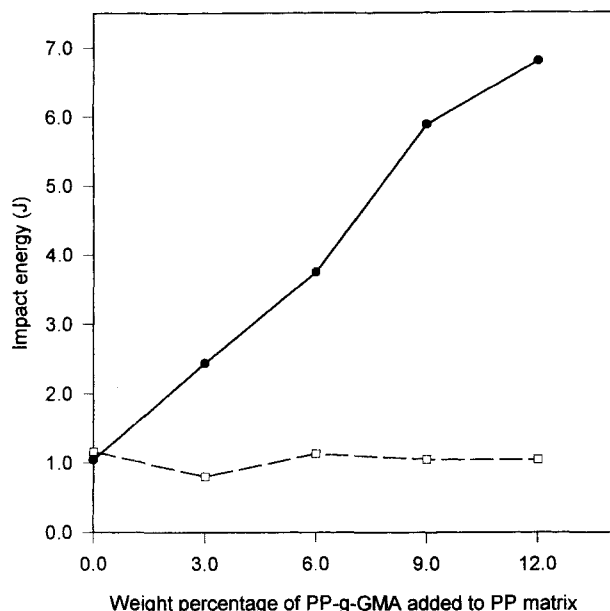


Fig. 13. Effect of total GMA concentration in matrix on impact energy of the PP/NBR blend by varying the weight percentage of the PP-g-GMA copolymer. The PP-g-GMA used has DG_g of 2.1 wt%. ● PP/PP-g-GMA/NBR blend. □ PP/PP-g-GMA blend with no NBR.

polymer (with similar MFI to the PP homopolymer) alone without NBR (Fig. 13). This means that the addition of the PP reactive copolymer alone in no way explains the increase in impact energy.

When PP reactive copolymers with different DG_g but similar MFI were added to the PP/NBR blends (at same weight percentage copolymer in PP matrix) the copolymers with higher DG_g compatibilized the blends better (Fig. 14). Since there was a higher total GMA concentration present in the matrix, more GMA functional groups were available for interfacial bonding for the same number of reactive copolymer molecules. Hence, the higher observed impact energy of the blend seems to be self-evident.

This suggests that it may be interesting to compare the compatibilizing performance of high DG_g copolymer and low DG_g copolymer on the basis of the same overall concentration of GMA functionality in the matrix. Two families of GMA grafted PP copolymers with a wide range of DG_g but similar MFI were synthesized, one with styrene grafts and one without. For each family of PP copolymers, the total GMA concentrations in the PP matrix were kept constant. It should be pointed out that since the DG_g 's differ largely, the weight percentages of the copolymers added to the matrix differ significantly as well. It was found that a large amount of low DG_g copolymer gives a higher impact energy for the PP/NBR blend than a small amount of high DG_g copolymer (Fig. 15 and 16). The impact energy of the matrix alone is, however, not affected, as illustrated by several reference PP/PP-g-GMA blends having the same matrix compositions as the corresponding PP/NBR blends. As the amount of copolymer was progressively reduced (DG_g getting

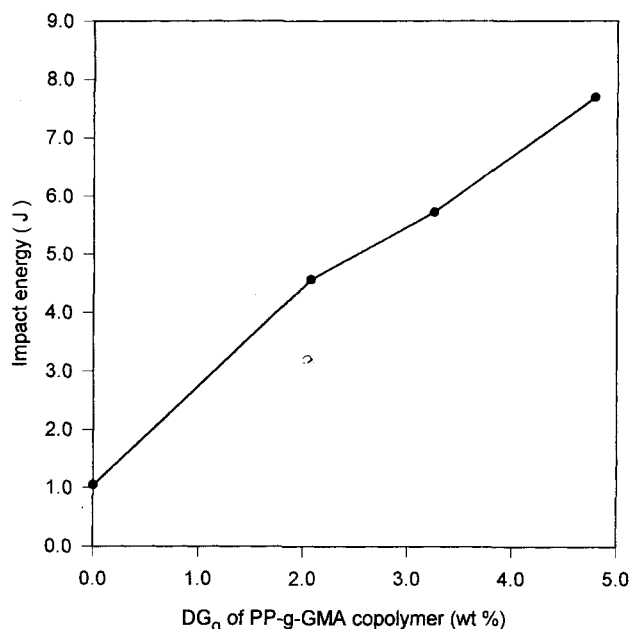


Fig. 14. Effect of total GMA concentration in matrix on impact energy of the PP/NBR blend by varying the DG_g of the PP-g-GMA copolymer. Weight percentage of the PP-g-GMA copolymer in matrix was kept constant at 9.0%.

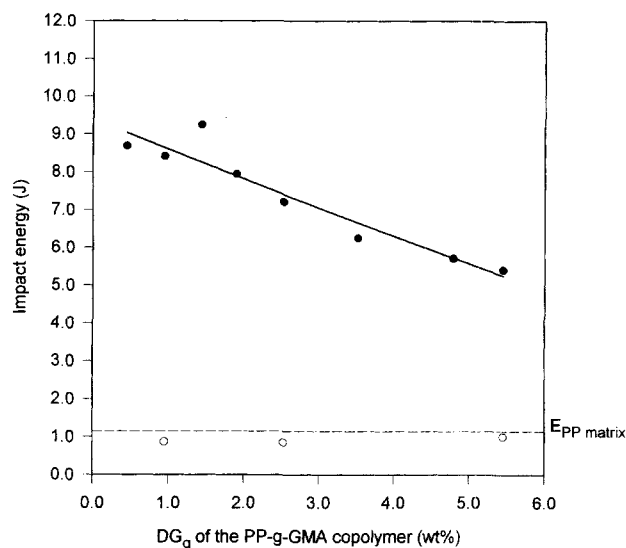


Fig. 15. Effect of DG_g of the PP-g-GMA copolymer on impact energy of the PP/NBR blend at a constant total GMA concentration in matrix. Total GMA concentration in matrix was kept constant at 0.14 wt%. MFI's of the matrix PP and all the PP-g-GMA copolymers are around 12.6. ● PP/PP-g-GMA/NBR blend. ○ PP/PP-g-GMA blend with no NBR.

even higher), the impact energy of the PP/NBR blend continued to decrease.

Such an effect is more obvious for the copolymers containing the styrene groups. The copolymers containing styrene groups seem to be less effective in compatibilizing the PP/NBR blends. Comparing the two curves in Fig. 15 and 16, it was found that reactive copolymers without styrene yielded somewhat higher impact energies than the corresponding copolymers

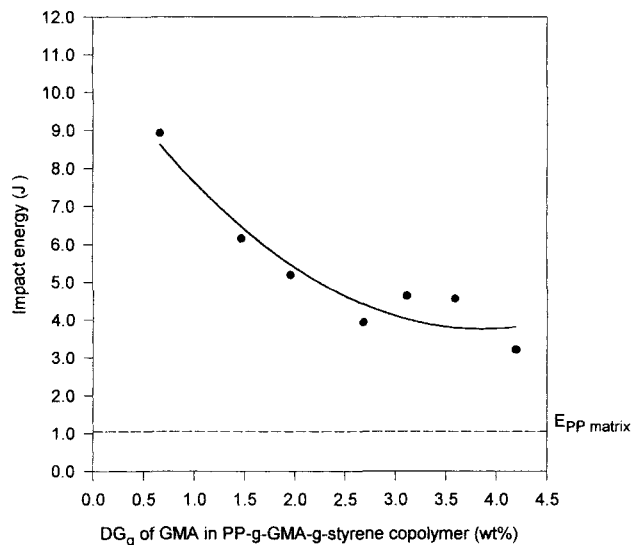


Fig. 16. Effect of DG_g of GMA of the PP-g-GMA-g-styrene copolymer on impact energy of the PP/NBR blend at a constant total GMA concentration in matrix. Total GMA concentration in matrix was kept constant at 0.30 wt%. MFT's of the matrix PP and all the PP-g-GMA copolymers are around 3.9.

with styrene at the same DG_g level. Note that the total GMA concentration in the matrix for the copolymers containing both GMA and styrene was higher than the concentration used for the GMA grafted copolymers. The decrease in impact strength may be caused by steric hindrance of the bulky benzene groups on the styrene unit. With styrene units randomly copolymerized with GMA units on the graft, the epoxy groups on the GMA units may not be able to access the carboxyl groups on the NBR as freely as those without styrene.

The direct influence of the molecular weight of these reactive copolymers was also studied. A family of PP-g-GMA copolymers having similar DG_g but varying molecular weight ($\bar{M}_n$) were compared. The weight percentage of copolymer was kept the same so that the total GMA concentration in the matrix is constant for all the blends. Figure 17 demonstrates that the effect of the copolymer molecular weights was quite strong. The impact energy of the blend decreased rapidly with decreasing molecular weight of the copolymer. It appears that the low molecular weight reactive copolymer does not have the same extent of entanglements in the matrix phase because of its shorter chain length.

Discussion of Effects of the Reactive Copolymer Characteristics

The effectiveness of a polymer additive to compatibilize an immiscible polymer blend can be attributed to how that polymer gets to the interface, how much of it is located at the interface, and what conformation it may have at the interface. It is generally assumed that the block copolymer and in-situ reactive graft copolymers stay at the interface if compatibilization occurs. With respect to physical compatibilization using block copolymers, these polymers usually have two blocks,

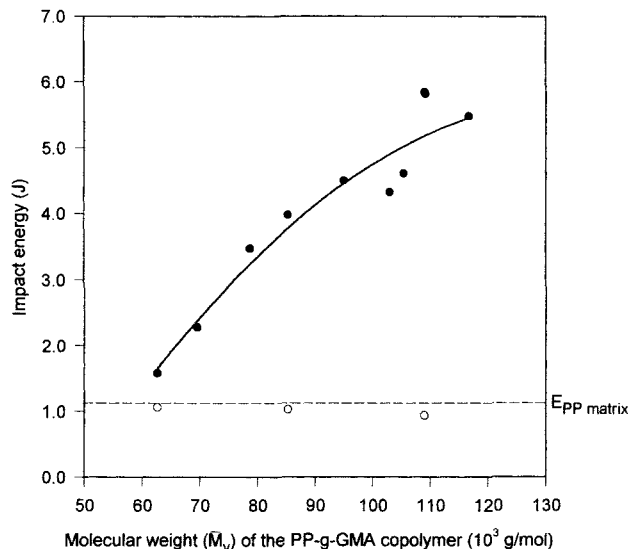


Fig. 17. Effect of molecular weight of the PP-g-GMA copolymer on impact energy of the PP/NBR blend. The matrix PP has a MW of 143,000 g/mol. DG_g of the PP-g-GMA copolymer are all around 2.7 wt%, and the total GMA concentration in matrix was kept constant at 0.27 wt%. ● PP/PP-g-GMA/NBR blend. ○ PP/PP-g-GMA blend with no NBR.

each is either identical to or miscible with one of the blend components and not the other. The mutual dissimilarity between each phase and the long blocks of the block copolymer are contributing factors for directing the copolymers to stay preferentially at the interface. While the central junction of the block copolymer is close to the interface, the blocks themselves are long enough to extend into each of the blend component phase and entangling to some extent (depending on the block molecular weight).

The situation for reactive graft copolymers has some similarities but also major differences. The graft copolymers normally have low graft contents (less than 5 wt%). Thus dissimilarity between such graft copolymers and the host homopolymer matrix is rather small, and the graft copolymers can be distributed fairly uniformly and stably in the host polymer phase. Since the PP-g-GMA and PP were blended first for three minutes, the reactive copolymer molecules should have been well distributed in the PP matrix before rubber was added. Thus, the reactive copolymer molecules have to somehow get to the interface to react with the carboxyl group in the dispersed rubber phase.

The graft chains are thermodynamically forced out from the bulk to the interface. During the course of dispersive and distributive mixing, fresh interfaces are generated by bulk melt movement (Fig. 18a) until an equilibrium morphology is established. Thus, there should be some reactive copolymer molecules steadily meeting the new interface. There is an excessive number of carboxyl groups from NBR and their reaction with the epoxy groups is rapid (41). Once a reactive copolymer molecule has arrived at the interface and reaction has occurred, the molecule is anchored to the

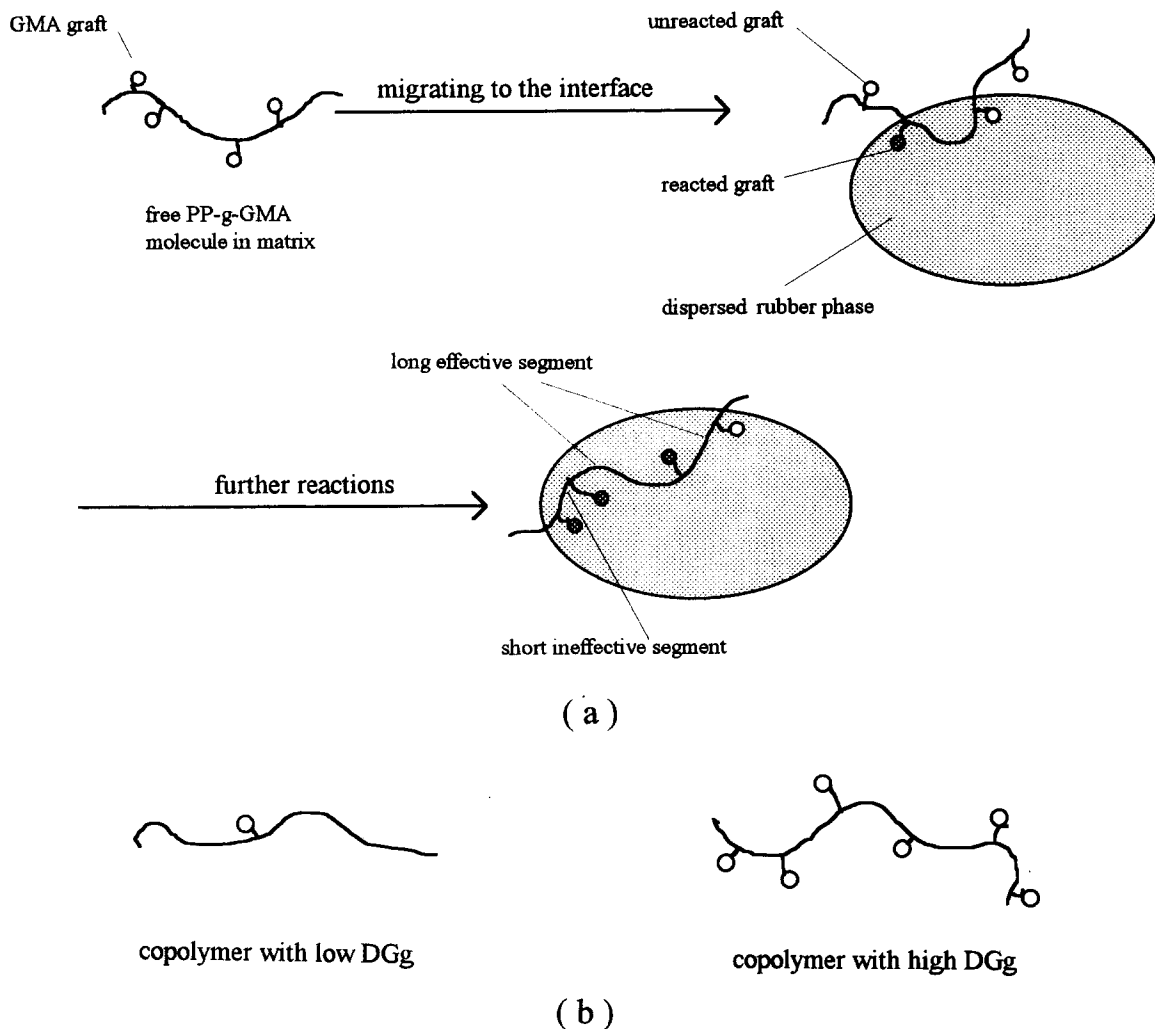


Fig. 18. Illustration of reactive compatibilization.

surface of the dispersed phase by the ester linkage formed. The movement of the anchored molecule is limited to the region at or near the interface. Hence, it is more likely for some of the remaining unreacted GMA grafts on this anchored molecule to react further with NBR than it is for the free reactive molecules still deeply buried in the PP phase. Therefore, multiple bonding between the same reactive copolymer and the rubber may be quite common among those which have reacted at the interface (bottom of Fig. 18a).

For the singly or multiply bound reactive PP copolymer molecules, their chains are now divided into two or more segments by the bonding joints at the surface of the rubber dispersed phase. Some of the segments may contain unreacted GMA grafts, which are probably still buried in the matrix. The segment length between bonding points would be expected to be a critical parameter related to how this reactive graft copolymer is entangled in the PP matrix. Hence, the molecular weight of the segment rather than the molecular weight of the whole graft copolymer molecule should be a controlling factor in interfacial bonding.

Kramer *et al.* (26, 27) studied the fracture mechanism at a polystyrene/poly(2-vinylpyridine) (PVP) interface reinforced with poly(styrene-*b*-2-vinylpyridine) block copolymer. They found that the minimum degree of polymerization of PVP for entanglement (N_{PVP}) is 255, below which the PVP block will pull out in a slow crack opening experiment. Computer simulation of entangled polymer chains using a local knot model estimated that the critical entanglement degree of polymerization for generic polymer molecules is about 230 (46), which translates into a critical entanglement molecular weight, MW_c , of about 9,700 g/mol for PP. Those segments attached at least at one end and having a molecular weight less than MW_c may not be sufficiently anchored in the matrix phase to resist a decohesion stress. Only segments with molecular weight greater than MW_c are capable of effectively entangling with the matrix and contribute to the overall compatibilizing performance.

The efficiency of reactive compatibilizer should depend primarily on the number of effective segments that are formed by the reaction of PP-g-GMA mole-

cules at the interface. The concentration of effective segments is related to the weight percentage of the copolymer in the matrix, the number-average molecular weight of the reactive copolymer (the maximum segment length if only end functionalized), and the DG_g of the copolymer (the number of times the molecular length can be partitioned). As the weight percentage of the same reactive copolymer is increased, the total number of copolymer molecules in the matrix increases, resulting in more reactive copolymer molecules arriving at the interface and a higher concentration of potentially effective segments. As a result, the impact energy of the blend increases accordingly (Fig. 13).

If the weight percentage of reactive copolymers is kept constant, and the MWs of these copolymers are similar, then the number of copolymer molecules present in the blend, and arriving at the interface, should be equivalent. Therefore, a higher DG_g means more grafts per molecule and higher total matrix phase GMA concentration (Fig. 14). Therefore, the impact energies of the blends compatibilized with progressively higher DG_g copolymers, at the same weight percentage in PP matrix, is increased. These results are in keeping with those obtained by Char *et al.* (30), who found that the concentration of functions groups was a dominant factor in reactive compatibilization.

In order to explain the results on Fig. 15 and 16, one can start with a sample calculation for the data points at the two extremes of the curve in Fig. 15. The molecular weights, $\bar{M}_v$, for the matrix and all the reactive copolymers are close to 150 kg/mol. Using a polydispersity value of 3.1, which was developed in the Experimental section, the $\bar{M}_n$ for both copolymers is estimated at about 50 kg/mol. For the lowest DG_g (0.45 wt%) copolymer, there are approximately 1.6 GMA units per chain on average. If it is assumed that each graft chain length is about 3–4 GMA monomer units, then every second graft copolymer molecule has a reactive graft. For the highest DG_g (5.5 wt%) copolymer, there are 19.5 GMA units, i.e., about 6 grafts per chain (Fig. 18b). In order to maintain uniformity of the GMA concentration in the matrix, twelve times higher weight percentage of the lowest DG_g copolymer was added as the highest DG_g copolymer. The efficiency of overall use of GMA grafts on the highest DG_g copolymer is lower, as there are fewer reactive copolymer molecules present in the blend. Therefore, for PP/NBR blends, a large amount of the lowest DG_g copolymer yields an impact energy about 60% higher than a small amount of the highest DG_g copolymer. A possible explanation for why the difference is not greater may lie in the limited overall interfacial surface area. If a large number of reactive copolymer molecules are already anchored on the surface of the rubber dispersed phase, it may be very difficult for other unreacted free copolymer molecules to access the interface.

In this case where the GMA concentration in the matrix is constant and the blends are prepared with progressively larger numbers of lower molecular

weight reactive copolymers, the impact strength of the blends decreased (Fig. 17). The number average molecular weight of these copolymers is estimated to range from 20 to 37 kg/mol. A probable explanation for the rapid drop in impact energy is the inability of the lower MW copolymers to entangle effectively after reaction with the NBR particles; hence, these molecules do not contribute significantly in improving the overall impact strength of the blends.

CONCLUSIONS

The glycidyl methacrylate (GMA) can be melt grafted onto polypropylene with via a free radical initiation but is accompanied by β -scission of the PP molecules. The extent of GMA grafting (DG_g) and the molecular weight of the functionalized PP copolymers can be controlled by carefully manipulating various reaction factors such as monomer concentration, initiator concentration, and reaction temperature as well as the molecular weight of the starting PP homopolymer. The addition of a second monomer to the reaction environment, namely styrene, permits one to increase the GMA grafting levels and reduce the extent β -scission.

The GMA grafted PP copolymer can effectively compatibilize the blends of PP homopolymer with acid-functionalized NBR rubber. Up to an eight-fold improvement in the impact energy of the PP/NBR blend has been obtained over those blends without the grafted copolymer. The compatibilizing capacity of the reactive copolymer is not exclusively dependent on the total reactive group concentration in the blend. The properties of the reactive copolymer are found to have a significant influence on the compatibilizing capacity. A large amount of moderately functionalized copolymer offers better compatibilization performance than a small amount of highly functionalized copolymer. Furthermore, reducing the molecular weight of the functionalized copolymer lowers its compatibilizing efficiency. The effect of adding styrene as a comonomer in the grafting process is to reduce the effectiveness of the GMA modified copolymers as blend compatibilizers.

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